

Where the searches go: a per-turn rubric audit of an over-searching GRPO search agent

Agentic-RL-CA — judge-reward-arms campaign, Phase 0/1 evidence report

2026-08-14 (SF time)

Abstract

Our step-75 GRPO search agent (Qwen3-8B-Base, ASearcher Base-35k, wiki-18/e5) spends ~ 5.4 searches per rollout at training temperature where a median of 3 suffice. To locate *where* the excess goes before changing the reward, we graded every search turn of 2,000 step-75 eval trajectories (34,125 turns) with an LLM judge (gpt-oss-120b) scoring five binary criteria per turn plus an evidence-readiness estimate p_t . Three findings. (1) The dominant error is a **repeated-query attractor**: first searches are nearly flawless (0.8% non-novel, 78% info-gaining), but from the third search onward 99.96% of queries are near-duplicates retrieving nothing new. (2) **Sufficiency arrives almost immediately and is then ignored**: 70.5% of trajectories reach $p_t \geq 0.9$ — on average at the *first* search — yet 99.9% of them keep searching, and two-thirds of all search turns occur after sufficiency. (3) Judge signals split sharply in predictive value: final readiness (AUC 0.74) and info-gain (0.69) discriminate success, while **well_formed** fails on 92.6% of turns *uniformly* across successes and failures (AUC 0.49), because the policy pastes the question verbatim as its query everywhere. We conclude that turn-level penalties should be gated on **novel/sufficiency**, that **well_formed** should be dropped from the penalty (a uniform tax duplicates the judge-free cost arm), and that p_t is a viable redistribution signal. Numbers here are an upper bound on the pathology: the audited rollouts are greedy-decoded and single-hop; the temperature-1.0 multi-checkpoint collection is in flight.

1 Motivation

The judge-reward-arms campaign asks whether rubric-based turn-level rewards can cut ~ 2 wasted searches per rollout at non-inferior accuracy (target $5.4 \rightarrow 3.5\text{--}4$ against the Base-35k T_{req} median of 3; required-turns study, 2026-08-04). Before training any arm, Phase 0/1 must establish (a) *what* the wasteful behaviour actually is, (b) whether the judge’s per-turn flags are precise enough to penalise it without collateral damage, and (c) whether readiness p_t carries enough signal to redistribute terminal reward across turns. This report is that evidence, computed on the cheapest available data: the existing greedy-decoded step-75 evaluation dump.

2 Method

Data. 2,000 trajectories from the step-75 GRPO checkpoint (`token_grpo_qwen3-8b-base_32turn_fast75_s0`) over NQ (1,000) and TriviaQA (1,000), greedy decoding, 32-turn horizon, top-5 wiki-18/e5 retrieval. 34,125 search turns judged; overall EM 0.503 (NQ 0.428, TriviaQA 0.577).

Judge. One gpt-oss-120b call per trajectory (temperature 0, JSON-schema constrained decoding; **0/2,000 parse fallbacks**), seeing the question, gold answers (privileged, training-time only), the final answer, and every search turn as **QUERY** + retrieved documents truncated to 600 characters each. The judge never sees the agent’s `<think>` text — criteria are scored behaviourally. Per turn t , five binary criteria:

Criterion	Pass condition
well_formed	concise keyword/entity query; not the question pasted verbatim, not garbled
relevant	targets an entity/relation on a minimal reasoning chain to a gold answer
novel	not a near-duplicate of an earlier query and not re-seeking already-retrieved facts
progression	uses a previously retrieved bridge fact to advance the hop chain (auto-pass on turn 1)
info_gain	this turn’s documents contain a new answer-relevant fact

plus readiness $p_t \in \{0, 0.1, \dots, 1.0\}$ — the probability a competent reader could state the gold answer from evidence retrieved in turns $1..t$ — and **sufficiency_turn**, the first t with $p_t \geq 0.9$.

3 Results

3.1 Over-search versus sufficiency

Under greedy decoding the policy issues **17.1 searches per rollout** (median 17; vs. ~ 5.4 at temperature 1.0). The judge says sufficient evidence arrives after **1.02 searches** on average, for the 70.5% of trajectories that ever reach $p_t \geq 0.9$. The two distributions barely overlap (Fig. 1): 99.9% of sufficient trajectories continue searching, issuing a mean of **16.1 post-sufficiency searches**; **66.3% of all search turns** occur after sufficiency. Under-retrieval exists but is the minority mode: trajectories that fail (EM= 0) end at mean $p = 0.53$, versus 0.91 for successes.

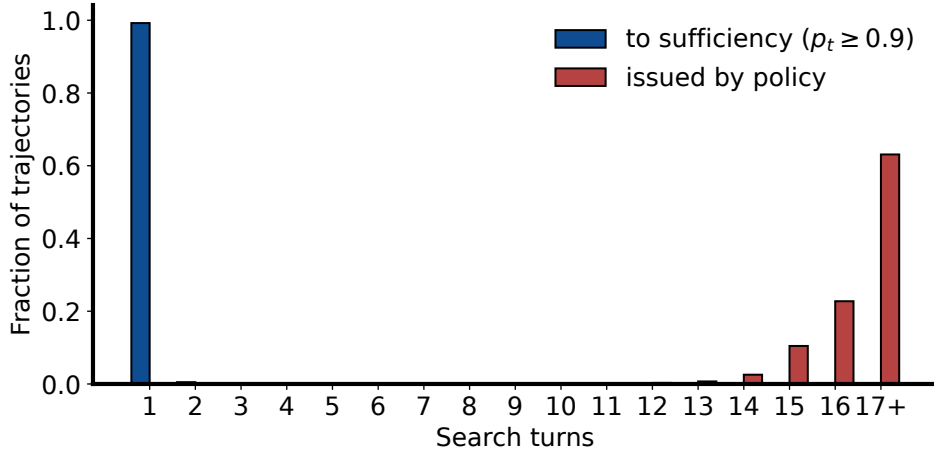


Figure 1: Searches needed (first $p_t \geq 0.9$; blue) versus searches issued (red), fraction of trajectories. Sufficiency is concentrated at one search on this single-hop slice; issued searches concentrate at the horizon.

3.2 The repeated-query attractor

Criterion failures are sharply position-dependent (Fig. 2, Table 1). The first search is nearly always sensible; by position ≥ 3 , **novel** fails on 99.96% of turns and **info_gain** on 99.98%: the policy re-issues the same or a near-duplicate query and re-retrieves the same documents. The mean longest consecutive non-novel run is **16.0 turns**; 99.95% of trajectories contain a run of ≥ 3 . By contrast, **well_formed** (92.6%) and **relevant** (10.8%) failure rates are flat in position: they measure query *style* (the policy pastes the question verbatim everywhere) and topicality, not the loop. The attractor is fully captured by **novel/info_gain/progression**.

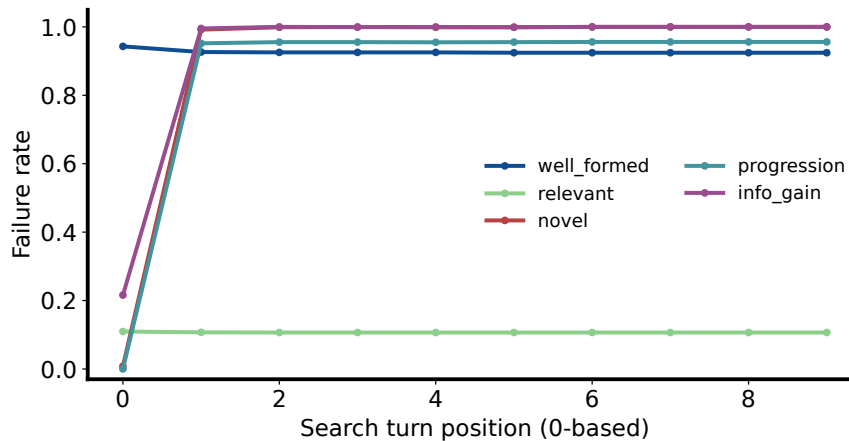


Figure 2: Criterion failure rate by search-turn position. The loop criteria (**novel**, **info_gain**, **progression**) jump from ≈ 0 to ≈ 1 after the first search; the style criteria (**well_formed**, **relevant**) are position-independent.

Criterion	fail@turn 0	fail@turns ≥ 3	fail, all turns
novel	0.008	0.9996	0.941
info_gain	0.216	0.9998	0.954
progression	0.000	0.956	0.900
well_formed	0.943	0.925	0.926
relevant	0.110	0.108	0.108

Table 1: Position dependence of criterion failures (34,125 turns).

3.3 Readiness dynamics

Mean p_t by position, split by outcome (Fig. 3): successful trajectories saturate readiness at the first search ($p \approx 0.9$ at position 0) and stay flat — subsequent searches add nothing the judge can see. Failed trajectories plateau near 0.5 and never recover, consistent with “retrieved the wrong entity’s page and kept re-retrieving it” rather than progressive refinement. $p_0 = 0$ for every trajectory.

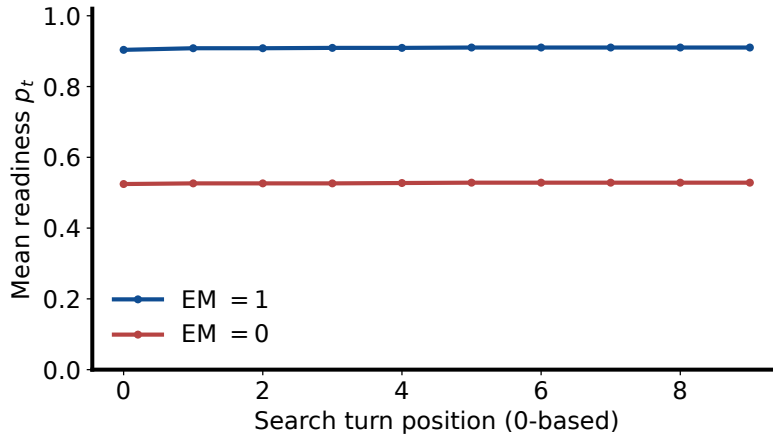


Figure 3: Mean readiness p_t by turn position and final outcome.

3.4 Which judge signals predict success?

Rank AUC of trajectory-level signals against EM (Fig. 4): p_{final} 0.74 and mean `info_gain` 0.69 carry real signal; `novel`/`progression`/`relevant` ≈ 0.56 are weakly informative individually on this single-hop slice; raw search count is similarly weak (0.56, fewer-is-better). `well_formed` is at chance (0.49): it fires near-uniformly on wins and losses alike.

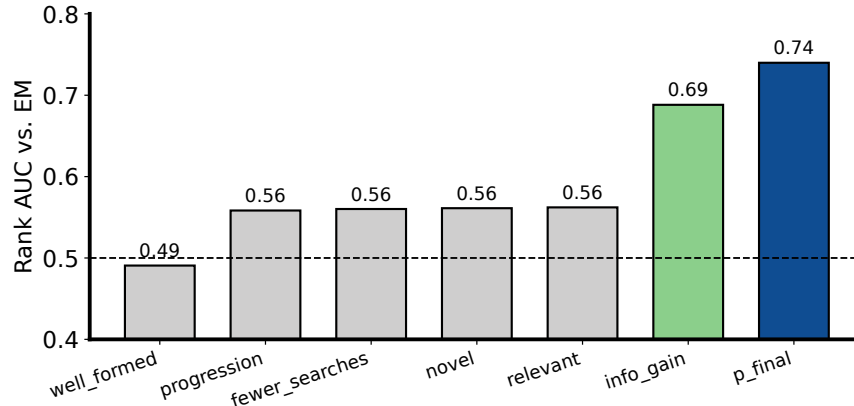


Figure 4: Predicting trajectory success (EM): rank AUC per judge signal. Dashed line = chance.

4 Implications for the reward arms

- **pen (asymmetric penalties).** The actionable penalty surface is `novel=0` (position-targeted, mechanism-aligned; §3.2) and $t > \text{ sufficiency_turn}$ (66% of turns; §3.1). **Drop well-formed from the penalty:** at 92.6% uniform failure and AUC 0.49 it is a flat per-search tax — behaviourally identical to the judge-free cost arm at vastly higher compute — and would dominate the summed penalty. Keep it as a logged canary.
- **redist (return-equivalent redistribution).** p_t saturates at the evidence-finding turn, so increments $p_t - p_{t-1}$ concentrate credit exactly where intended (the turn that found the evidence, not the spin). AUC 0.74 for p_{final} supports the signal; formal calibration (reliability of p vs. empirical solve rate) runs on the temperature-1.0 collection.
- **cost (judge-free control).** The attractor gives even a blind per-search cost a large target; the weak search-count AUC (0.56) is the a-priori case that *which* searches get cut matters — i.e. why pen may retain accuracy better than cost.
- **Judge operations.** JSON-schema mode: 0/2,000 fallbacks; ~45 min for 2,000 trajectories on 4×H100 at concurrency 32 — comfortably inside the ~6M input-token/step budget for training-time judging (1,280 trajectories/step).

5 Limitations

- **Greedy decoding.** Repetition attractors are amplified at temperature 0; the training-time profile is ~5.4 searches/rollout at temperature 1.0. Magnitudes here are an upper bound; the position-dependence *shape* is the transferable finding. The temperature-1.0 collection from baseline checkpoints 25/50/75 (in flight) is the training-matched replication.
- **Single-hop slice.** The 2,000-trajectory cap covers NQ + TriviaQA only; **progression** is rarely exercised and sufficiency lands on the first search partly by task construction. Multi-hop criterion behaviour (HotpotQA, 2Wiki, Musique) is unmeasured here.
- **Judge as ground truth.** Criterion “failure rates” are judge assessments; the Phase-1 launch gate includes a manual precision spot-check (~100 trajectories, $\geq 80\%$ precision per criterion) before any flag earns a gradient.
- **Sufficiency $\neq$ answerability.** $p_t \geq 0.9$ is the judge’s evidence estimate given gold answers; a policy cannot see p_t and must learn stopping from the reward.

6 Reproducibility

Branch `judge-reward-arms` (repo `Agentic-RL-CA`), report directory `docs/reports/2026-08-14_rubric_error_patterns/`. Figures pair with their data as `assets/fig_*.pdf` + `assets/fig_*.csv`.

- Rubric judging (4×H100 worker, serves `gpt-oss-120b` + judges the dump):
`DUMPS="grp0_s75=$HLOG/p8b_eval_grp0_s75/rollout_eval_grp0_s75.jsonl.zst"`
`GOLDSET=.../asearcher_eval.parquet MAX_TRAJS=2000 bash scripts/asearcher/p8b_rubric_judge_worker.sh`
- This paper’s statistics and figures: `uv run make_paper_figs.py`; the HTML companion report: `index.html` (same directory).